The LKT augmented GRU: predict first, then update memory

Forward-only: 3 GRU layers × 32 units | 21 predictor coordinates | 20,977 parameter entries

A Network at one trial t, for one participant

KC identity
72 known IDs + 1 reserved row

Learned KC embedding
16 coordinates

5 numeric coordinates
log(1 + student time gap)
intercept + 3 LKT predictors
training-only standardization

Concatenate 5 + 16
Input dropout 0.1
x(t): 21 coordinates

x(t)

Concatenate
[h3(t−1), x(t)]
32 + 21 = 53

Linear 53 → 32
ReLU
Head dropout 0.1

Linear 32 → 1
One logit

Sigmoid + clipping
p(correct at trial t)
ε = 10⁻⁵

1 Predict from inputs and PREVIOUS memory

Previous TOP-layer memory goes to the head

h1(t−1)
32 previous values

h2(t−1)
32 previous values

h3(t−1)
32 previous values

2 Observe the answer, then update

Observed answer y(t)
0 = incorrect; 1 = correct

Concatenate
[x(t), y(t)]
21 + 1 = 22

h1(t): 32

h2(t): 32

GRU layer 1
32 units
Own state

Dropout
0.1

GRU layer 2
32 units
Own state

Dropout
0.1

GRU layer 3
32 units
Own state

h3(t)
32 values

Retain ALL three states for the next trial: [h1(t), h2(t), h3(t)] = 3 × 32 values

Each new participant starts with zero states. New memory h3(t) can affect the NEXT prediction, p(t+1).

Dropout locations are shown for training; dropout is disabled during evaluation and continuation.

B Inside one GRU layer: the exact Torch reset/update equations

Every layer has separate weights and biases. u(t) has 22 coordinates in layer 1 and 32 in layers 2 and 3.

Layer input
u(t)

Separate learned affine maps
for g in {reset r, update z, new n}

a_g = W_ig u(t) + b_ig

c_g = W_hg h(t−1) + b_hg

Each a_g and c_g has 32 values

Previous state
h(t−1): 32

UPDATE gate
z(t) = sigmoid(a_z + c_z)

z(t)

RESET gate
r(t) = sigmoid(a_r + c_r)

r(t)

CANDIDATE state
n(t) = tanh(a_n + r(t) ⊙ c_n)

n(t)

Mix candidate and old state

h(t) = (1−z(t)) ⊙ n(t)
+ z(t) ⊙ h(t−1)

New state
h(t): 32

Old state: identity carry path

⊙ means elementwise multiplication. The reset gate multiplies c_n INCLUDING its recurrent bias, as in torch::nn_gru.

LKT columns: intercept, student logitdec, KC PPES, and student–KC logitdec. The centered intercept coordinate is zero.

Training also compares p(t) with y(t) using binary cross-entropy. Continuation updates hidden states while weights stay fixed.

Reported probabilities are clipped to [ε, 1−ε], with ε = 10^{-5}. Training loss uses the un-clipped logits.

Source: R/lkt-model-gru.R and R/lkt-gru-prediction.R. Dimensions and equations verified against augmented_g003_seed1.